CHOW-TYPE SECTION THROUGH A GERBE OVER A DENSE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2230-2251 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2253-2456, the complete author proof (202 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_stacks-stacks-more-morphisms.json (pb=8142, stmt 2230-2251, proof 2253-2456).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: the initial remark reduction is 'Some details omitted' - assumption bookkeeping, non-load-bearing (artin #2 precedent).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-finite-cover-factor: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2201–2207. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-flatten-stack: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2116–2124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09YC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2456–2456. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CPI: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2456–2456. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 084N: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2456–2456. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06QI: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2456–2456. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CPR: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2456–2456. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 050J: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2456–2456. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CPL: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2456–2456. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : \mathcal{X} \to Y$ be a morphism from an algebraic stack to an algebraic space. Let $V \subset Y$ be an open subspace. Assume*

(1) *f is separated and of finite type,*
(2) *Y is quasi-compact and quasi-separated,*
(3) *V is quasi-compact, and*
(4) *$\mathcal{X}_V$ is a gerbe over V.*

Then there exists a commutative diagram

$$\begin{array}{ccccc} \overline{Z} & \xleftarrow{j} & Z & \xrightarrow{h} & \mathcal{X} \\ & \searrow^{\overline{g}} & \downarrow{\scriptstyle g} & \swarrow_{f} & \\ & & Y & & \end{array}$$

with j an open immersion, $\overline{g}$ and h proper, and such that $|V|$ is contained in the image of $|g|$.

Proof. Suppose we have a commutative diagram

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \longrightarrow & Y \end{array}$$

and a quasi-compact open $V' \subset Y'$, such that $Y' \to Y$ is a proper morphism of algebraic spaces, $\mathcal{X}' \to \mathcal{X}$ is a proper morphism of algebraic stacks, $V' \subset Y'$ maps surjectively onto V, and $\mathcal{X}'_{V'}$ is a gerbe over V'. Then it suffices to prove the lemma for the pair $(f' : \mathcal{X}' \to Y', V')$. Some details omitted.

Overall strategy of the proof. We will reduce to the case where the image of f is open and f has a section over this open by repeatedly applying the above remark.

Each step is straightforward, but there are quite a few of them which makes the proof a bit involved.

Using Limits of Spaces, Proposition 09YC we reduce to the case where Y is a scheme. (Let $Y' \to Y$ be a finite surjective morphism where Y' is a scheme. Set $\mathcal{X}' = \mathcal{X}_{Y'}$ and apply the initial remark of the proof.)

Using Lemma lemma-flatten-stack (and Morphisms of Stacks, Lemma 06QI to see that a gerbe is flat and locally of finite presentation) we reduce to the case where f is flat and of finite presentation.

Since f is flat and locally of finite presentation, we see that the image of $|f|$ is an open $W \subset Y$. Since $\mathcal{X}$ is quasi-compact (as f is of finite type and Y is quasi-compact) we see that W is quasi-compact. By Lemma lemma-finite-cover-factor we can find a finite surjective morphism $g : Y' \to Y$ such that $g^{-1}(W) \to Y$ factors Zariski locally through $\mathcal{X} \to Y$. After replacing Y by Y' and $\mathcal{X}$ by $\mathcal{X} \times_Y Y'$ we reduce to the situation described in the next paragraph.

Assume there exists $n \geq 0$, quasi-compact opens $W_i \subset Y$, $i = 1, \ldots, n$, and morphisms $x_i : W_i \to \mathcal{X}$ such that (a) $f \circ x_i = \mathrm{id}_{W_i}$, (b) $W = \bigcup_{i=1,\ldots,n} W_i$ contains V, and (c) W is the image of $|f|$. We will use induction on n. The base case is $n = 0$: this implies $V = \emptyset$ and in this case we can take $\overline{Z} = \emptyset$. If $n > 0$, then for $i = 1, \ldots, n$ consider the reduced closed subschemes Y_i with underlying topological space $Y \setminus W_i$. Consider the finite morphism

$$Y' = Y \amalg \coprod\nolimits_{i=1,\ldots,n} Y_i \longrightarrow Y$$

and the quasi-compact open

$$V' = (W_1 \cap \ldots \cap W_n \cap V) \amalg \coprod_{i=1,\ldots,n} (V \cap Y_i).$$

By the initial remark of the proof, if we can prove the lemma for the pairs

$$(\mathcal{X} \to Y, W_1 \cap \ldots \cap W_n \cap V) \quad \text{and} \quad (\mathcal{X} \times_Y Y_i \to Y_i, V \cap Y_i), \quad i = 1, \ldots, n$$

then the result is true. Here we use the settheoretic equality $V = (W_1 \cap \ldots \cap W_n \cap V) \cup \bigcup_{i=1,\ldots,n} (V \cap Y_i)$. The induction hypothesis applies to the second type of pairs above. Hence we reduce to the situation described in the next paragraph.

Assume there exists $n \geq 0$, quasi-compact opens $W_i \subset Y$, $i = 1, \ldots, n$, and morphisms $x_i : W_i \to \mathcal{X}$ such that (a) $f \circ x_i = \mathrm{id}_{W_i}$, (b) $W = \bigcup_{i=1,\ldots,n} W_i$ contains V, (c) W is the image of $|f|$, and (d) $V \subset W_1 \cap \ldots \cap W_n$. The morphisms

$$T_{ij} = Isom_{\mathcal{X}}(x_i|_{W_i \cap W_j \cap V}, x_j|_{W_i \cap W_j \cap V}) \longrightarrow W_i \cap W_j \cap V$$

are surjective, flat, and locally of finite presentation (Morphisms of Stacks, Lemma 0CPR). We apply Lemma lemma-finite-cover-factor to each quasi-compact open $W_i \cap W_j \cap V$ and the morphisms $T_{ij} \to W_i \cap W_j \cap V$ to get finite surjective morphisms $Y'_{ij} \to Y$. After replacing Y by the fibre product of all of the Y'_{ij} over Y we reduce to the situation described in the next paragraph.

Assume there exists $n \geq 0$, quasi-compact opens $W_i \subset Y$, $i = 1, \ldots, n$, and morphisms $x_i : W_i \to \mathcal{X}$ such that (a) $f \circ x_i = \mathrm{id}_{W_i}$, (b) $W = \bigcup_{i=1,\ldots,n} W_i$ contains V, (c) W is the image of $|f|$, (d) $V \subset W_1 \cap \ldots \cap W_n$, and (e) x_i and x_j are Zariski locally isomorphic over $W_i \cap W_j \cap V$. Let $y \in V$ be arbitrary. Suppose that we can find a quasi-compact open neighbourhood $y \in V_y \subset V$ such that the lemma is true for the pair $(\mathcal{X} \to Y, V_y)$, say with solution $\overline{Z}_y, Z_y, \overline{g}_y, g_y, h_y$. Since V is quasi-compact, we can find a finite number $y_1, \ldots, y_m$ such that $V = V_{y_1} \cup \ldots \cup V_{y_m}$.

Then it follows that setting

$$\overline{Z} = \coprod \overline{Z}_{y_j}, \quad Z = \coprod Z_{y_j}, \quad \overline{g} = \coprod \overline{g}_{y_j}, \quad g = \coprod g_{y_j}, \quad h = \coprod h_{y_j}$$

is a solution for the lemma. Given y by condition (e) we can choose a quasi-compact open neighbourhood $y \in V_y \subset V$ and isomorphisms $\varphi_i : x_1|_{V_y} \to x_i|_{V_y}$ for $i = 2, \ldots, n$. Set $\varphi_{ij} = \varphi_j \circ \varphi_i^{-1}$. This leads us to the situation described in the next paragraph.

Assume there exists $n \geq 0$, quasi-compact opens $W_i \subset Y$, $i = 1, \ldots, n$, and morphisms $x_i : W_i \to \mathcal{X}$ such that (a) $f \circ x_i = \mathrm{id}_{W_i}$, (b) $W = \bigcup_{i=1,\ldots,n} W_i$ contains V, (c) W is the image of $|f|$, (d) $V \subset W_1 \cap \ldots \cap W_n$, and (f) there are isomorphisms $\varphi_{ij} : x_i|_V \to x_j|_V$ satisfying $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$. The morphisms

$$I_{ij} = Isom_{\mathcal{X}}(x_i|_{W_i \cap W_j}, x_j|_{W_i \cap W_j}) \longrightarrow W_i \cap W_j$$

are proper because f is separated (Morphisms of Stacks, Lemma 0CPL). Observe that φ_{ij} defines a section $V \to I_{ij}$ of $I_{ij} \to W_i \cap W_j$ over V. By More on Morphisms of Spaces, Lemma 0CPI we can find V-admissible blowups $p_{ij} : Y_{ij} \to Y$ such that s_{ij} extends to $p_{ij}^{-1}(W_i \cap W_j)$. After replacing Y by the fibre product of all the Y_{ij} over Y we get to the situation described in the next paragraph.

Assume there exists $n \geq 0$, quasi-compact opens $W_i \subset Y$, $i = 1, \ldots, n$, and morphisms $x_i : W_i \to \mathcal{X}$ such that (a) $f \circ x_i = \mathrm{id}_{W_i}$, (b) $W = \bigcup_{i=1,\ldots,n} W_i$ contains V, (c) W is the image of $|f|$, (d) $V \subset W_1 \cap \ldots \cap W_n$, and (g) there are isomorphisms $\varphi_{ij} : x_i|_{W_i \cap W_j} \to x_j|_{W_i \cap W_j}$ satisfying

$$\varphi_{jk}|_V \circ \varphi_{ij}|_V = \varphi_{ik}|_V.$$

After replacing Y by another V-admissible blowup if necessary we may assume that V is dense and scheme theoretically dense in Y and hence in any open subspace of Y containing V. After such a replacement we conclude that

$$\varphi_{jk}|_{W_i \cap W_j \cap W_k} \circ \varphi_{ij}|_{W_i \cap W_j \cap W_k} = \varphi_{ik}|_{W_i \cap W_j \cap W_k}$$

by appealing to Morphisms of Spaces, Lemma 084N and the fact that $I_{ik} \to W_i \cap W_j$ is proper (hence separated). Of course this means that (x_i, φ_{ij}) is a desent datum and we obtain a morphism $x : W \to \mathcal{X}$ agreeing with x_i over W_i because $\mathcal{X}$ is a stack. Since x is a section of the separated morphism $\mathcal{X} \to W$ we see that x is proper (Morphisms of Stacks, Lemma 050J). Thus the lemma now holds with $\overline{Z} = Y$, $Z = W$, $\overline{g} = \mathrm{id}_Y$, $g = \mathrm{id}_W$, $h = x$. $\square$